

OPENFLEX continuation: locating delayed amplification in the carried financial state

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Finding

The delayed response is carried mainly by the joint financial state in this fixture. At t20 the financial positions, wealth, rates and reciprocal quotes together contribute approximately **1.40055** to the one-period quote gain; other real and diagnostic lag states contribute **−0.299406**, producing the observed **1.101145** gain. The same decomposition at t11 gives about $1.393639 - 0.315914 = \mathbf{1.077725}$. Both signed disturbance sizes agree.

These are derivatives along the previously solved early spending-pulse paths, not eigenvalues, independent policy effects or a proof of a unique causal loop. The joint financial direction preserves the selected lagged financial, portfolio and conversion constraints to measured precision. Splitting it into wealth, quote and asset-position directions does not. The grouping therefore matters for interpretation, even though all terms add to the same total.

OPENFLEX remains excluded. No new calibration, behavioural equation, accepted cache or manuscript change was made. This authorized continuation advances Task 3 after the earlier bounded exclusion decision; it does not retroactively turn that decision into a successful stabilization.

Exact inputs and experiment

Parent: theta=1, sigma=0, Xi-consistent revised OPENFLEX, `output/scenarios/arms/revised_openflex/runs/b` SHA-256 80dc540620c1ddd971c2c4a68fa3a95959140518445521ef9dba6789ae257471. All 21 retained launch sources and saved options validate through `R/dynamics/arm_context.R`. Input full states are the signed $\pm 1e-4$ and $\pm 1e-5$ branches in `leeds_diagnosis/pulses_full_01/`. Those early branches have separate financial/cash-budget/domain receipts. Their first fully unforced current period is t11. The t10 state still records the cancelling input; the complete lag tangent is retained rather than incorrectly requiring that lagged input to be zero at t11.

No new economic trajectory or stabilization candidate was simulated. At t11 and t20 we settle the current equations with a fixed quote, changing only the lag input along the measured pulse tangent. All current exogenous input differences are zero. Each inner solve uses a $1e-12$ settlement tolerance and at least 130 sweeps. We also differentiate the fixed-quote gap with respect to the current quote itself. The inherited redundant EU bill residual is not erased.

Implicit reconstruction and precision

Let $G(q, s_previous, A_previous)$ be the settled foreign-bill clearing gap. Along the actual solved tangent, the implicit quote response is

$$dq = -\frac{G_s ds_{previous} + G_A dA_{previous}}{G_q}.$$

We measure centered differences at both disturbance sizes. The detailed partition covers all lag coordinates without overlap; its CSV names every coordinate. A second, independently evaluated partition combines positions, wealth, returns/expectations and reciprocal quotes into one carried financial direction. Each partial settles every current equation other than quote clearing. It is a derivative diagnostic; the partial point need not be a fully feasible prior economic history.

Period	Size	ObservedGain	Reconstructed	ScaledError	Additivity
11	0.0001	1.07772	1.07772	4.24605e-10	2.19305e-10
20	0.0001	1.10114	1.10114	2.93586e-09	1.01524e-09
11	1e-05	1.07772	1.07772	8.91947e-10	1.27889e-09
20	1e-05	1.10114	1.10114	8.0264e-11	6.83094e-11

For the detailed partition, maximum scaled reconstruction error is 2.04063e-09; the largest gain contribution change between sizes is 8.17868e-09. The direct coupled evaluation also agrees with the summed detailed financial terms to measured numerical precision. This agreement checks the fixed-quote implicit reconstruction and partition additivity, rather than presupposing that a supply/demand log decomposition identifies causes.

Period	Group	GainContribution
11	Other real/diagnostic states	-0.315914
11	Joint financial state	1.39364
11	Carried technology	0
20	Other real/diagnostic states	-0.299406
20	Joint financial state	1.40055
20	Carried technology	0

The income/return/expectation group has a zero derivative for this particular pulse and saved sigma=0 configuration; this does not make income dynamics irrelevant in general. Current incomes, consumption, prices and trade still respond while the inner equations settle. Carried technology differences are zero; within-period Armington substitution remains active.

Why the separate terms are not independent interventions

At t20, the detailed bookkeeping attribution is:

Group	GainContribution
Other real/diagnostic states	-0.299406
Financial positions	2.51521

Group	GainContribution
Income/returns/expectations	0
Household wealth	-1.18693
Reciprocal quotes	0.0722715
Carried technology	0

Taken separately, these directions violate lagged household wealth, fixed portfolio shares or issuer/holder currency conversion. The split constraint audit checks reciprocity, both household financial net-worth identities, both bank balance sheets, both bill markets, four foreign conversions and eight portfolio-share identities. The full pulse tangent satisfies their derivatives within $1.81929\text{e-}09$ normalized units per unit pulse. The financial-position and quote splits each have a maximum derivative near **0.895** at t20. Combining positions and wealth alone still leaves the conversion constraint violated; the reciprocal quotes must move with them. The joint financial direction passes these selected constraints within $1.81929\text{e-}09$ for the full tangent; its own constraint rows are retained separately in the coupled audit.

These are selected constraints, not a complete independent real-capital/net-worth matrix or proof that the partial lag states have feasible preceding histories. Attribution also depends on coordinate choice. It would be incorrect to read the +2.515 position term as a separately implementable destabilizing policy and the -1.187 wealth term as a separately implementable stabilizer.

What this makes testable next

The static clearing ratio alone is insufficient to explain the delayed gain: G_q is approximately -2.295 at t11 and -2.176 at t20, while the lagged financial state carries the net positive response. The next useful diagnosis is a reduced set of independent lag coordinates that reconstructs financial positions, wealth and currency conversion together, then evaluates the solved transition on that set. Bill supply, household accumulation and valuation must be traced as one consistent loop. A full transition spectrum and independent real-capital/net-worth audit remain unresolved.

No further parameter sweep is justified by this derivative attribution alone. A separately specified behavioural/institutional candidate would still need root, accounting, financial-domain, stationarity and signed-attraction gates. The missing OPENECO workbook/native fixtures remain a separate route.

Reproducibility and retained failures

Scripts: `R/dynamics/openflex_lag_channels.R` and `R/dynamics/openflex_lag_constraint_check.R`.
Current outputs:

- `output/calibration/openflex_attack/leeds_diagnosis/lag_channels_03/`
- `output/calibration/openflex_attack/leeds_diagnosis/lag_channels_coupled_01/`

Source snapshots, bundle hashes, saved options, elapsed runtimes, complete partitions, derivative tables and validation receipts are retained. The initial detailed launches failed on a signed filename format and an incorrect lagged-input assertion before numerical evaluation; the first split-constraint harness lost normalization names. Failed receipts remain in their original directories. These are harness failures, not scientific evidence against a closure. No failed or successful directory was repurposed.

The detailed evaluation took 83.719 seconds; the coupled evaluation took 55.896 seconds. All 200 protected files, including 104 accepted caches, revalidate unchanged. This continuation remains uncommitted; nothing was published or sent.